CNuvS Ex. 9 - Jan Lindauer, Paul Feidieker

Task 5

(a) Kendall's queueing notation:

- **A**: Arrival process
- **S**: Service process
- **m**: Number of servers
- **N**: places in the system (bounded queue length), if not given, then assumed inf
- **K**: Population Size
- **SD**: Queue discipline

A and **S** are noted as follows:

- **M**: Exponential process (Markovian)
- **D**: Deterministic
- **G**: General

[1, p. 36]

(b) The notation is structured as A/S/m, where each component gives a specific rule about how the system behaves.

The first **M** in M/M/1 represents **A**, an exponential process, specifically a Markovian.
The second **M** represents **S**, also an exponential process, specifically a Markovian.
The **1** stands for **m**, the number of servers.
[1, p. 36, 37, 44]

(Parameters for c - g)
Arrival Rate: 25 req/sec
Service Time ($E(S)$): $\frac{1}{\mu} = 0.02$ seconds
Service Rate (μ): $\mu = 50$ req/sec

(c) The average system time represents the total time a request spends in the system (waiting time + service time).
Formula:

$$W = \frac{1}{\mu - \lambda}$$

Calculation:

$$W = \frac{1}{50 - 25} = \frac{1}{25} = 0.0400 \quad \text{seconds}$$

[1, p. 40]

(d) Utilization measures the fraction of time that the server is busy handling requests.
Formula:

$$\rho = \frac{\lambda}{\mu}$$

Calculation:

$$\rho = \frac{25}{50} = 0.5000$$

[1, p. 39]

(e) This represents the total expected number of requests currently being served and waiting in the queue.
Formula:

$$L = \frac{\lambda}{\mu - \lambda} = \frac{\rho}{1 - \rho}$$

Calculation:

$$L = \frac{25}{50 - 25} = \frac{25}{25} = 1.0000 \text{ request}$$

[1, p. 40]

(f) The probability of having exactly n customers/requests in a steady-state M/M/1 system follows a geometric distribution.
Formula:

$$P_n = (1 - \rho)\rho^n$$

Calculation for $n = 4$:

$$P_4 = (1 - 0.5) \times (0.5)^4 = 0.5 \times 0.0625 = 0.0313$$

[1, p. 39]

(g) The probability that the server is completely idle is $P_0 = 1 - \rho$. Over a time interval T, the expected total idle duration is the idle probability multiplied by T.
Formula:

$$I = (1 - \rho) \times T$$

Calculation:
First, convert the time interval to seconds: $T = 6 \text{ minutes} \times 60 \text{ seconds/minute} = 360 \text{ seconds}$.

$$I = (1 - 0.5) \times 360 = 180.0000 \text{ seconds}$$

Task 6

The maximum segment size (MSS) is 4KB and the maximum congestion window is 64KB. Therefore the maximum congestion window can contain

$$\frac{64}{4} = 16 \text{ MSS}$$

1. A timeout event reduces the congestion window to 1 MSS. The AIMD mechanism increments the congestion window by 1 MSS every round-trip time (RTT). Therefore it takes 15 RTT to return back to the maximum congestion window. This takes $80 \text{ msec} * 15 \text{ RTT} = 1200 \text{ msec} = 1.2 \text{ sec}$.

2. The triple duplicate ACKs event only halves the congestion window to a size of 8 MSS. To return to the maximum size for the congestion window it therefore takes 8 RTT. This means it takes $80 \text{ msec} * 8 \text{ RTT} = 640 \text{ msec} = 0.64 \text{ sec}$.

Sources

[1] Prof. Dr. Marco Zimmerling, “CNuvS 05: Queueing Theory.” 2026.